

Omer Pinto

☎ (+972) 54-534-7868 | ✉ omerp2006@gmail.com | 🌐 Omer-Pinto | 📄 omer-pinto-a940a8122

Summary & Current Focus

- Senior Software Engineer with **15+ years** building **backend systems, database internals, & distributed platforms**.
- Strong foundation in **algorithms, performance tuning & systems architecture across on-prem & cloud platforms**.
- Currently developing **Python** microservices and orchestration systems at BigArch while actively expanding into **AI Engineering**.
- Seeking **AI Engineering roles** focused on intelligent products and **LLM-powered applications**.

Experience

BigArch

SENIOR SW ENGINEER

2024 - Present

- Backend development of a large-scale **graph orchestration platform** for HPC with **Python, MongoDB, RabbitMQ & microservices**.
- Architect and deploy containerized services in private-cloud environment using **Kubernetes** and **Docker**.
- Achieved **60% performance gains** through system-wide optimization, transforming strained system into stable, scalable platform.
- Lead system-wide architectural refactors executed with **zero downtime** and full service continuity.
- **AI Initiatives:** Evaluated multiple LLMs in air-gapped system. Built custom shim server enabling LLM tool integration.
- Drive AI-assisted development adoption through practical MVPs & team enablement (More on AI work & self-study in Projects section)

Microsoft

SENIOR SW ENGINEER

2023 - 2024

- Engineer in the **vulnerability management** group, developing components in the endpoint security pipeline.
- Built **C# microservices** and data pipelines using Azure services (Blobs, Queues, CosmosDB, KustoDB).

Yahoo

PRINCIPAL SW ENGINEER

2022 - 2023

- Contributor to the **native ads** platform, developing **Java** services and **Hadoop/Spark** ETL pipelines.
- Partnered with research team to implement **ML/DL-driven ad-scoring algorithms** for multi-variable ranking system.
- Managed codebase migration and team handoff following **project cancellation**.

qSpark

SENIOR SW ENGINEER

2021 - 2022

- Delivered **ultra-low-latency trading** infrastructure. Implemented **exchange protocols** (execution & market data) using **modern C++ with template meta-programming**.
- Designed **scalable** subsystem enabling 3rd-party **trading algorithms** integration.
- Handled production issues with **global exchanges** and conducted market-structure research (SEC/FINRA rules).

Toga Networks

HANDS-ON ARCHITECT

2019 - 2021

- Worked across multiple projects in the **cloud and IoT R&D** group, driving architecture and implementation.
- Performed **market and technology research** and delivered high-level designs for innovative systems.
- Led research & design of semantic analytics engine for IoT data, including reasoning capabilities across data model and query layers.
- Developed a log-analysis algorithm using **exponential backoff**, improving system performance.

SiSense

SENIOR ENGINEER, TECH LEAD & ARCHITECT, CTO OFFICE

2014 - 2018

- **Hand-picked by CTO** for his own private team researching next-gen analytics & developing features on open-source DB project.
- Architected and built cross-platform **ETL query engine from scratch** as sole developer; recruited and led team as tech lead.
- Engine optimized for **low RAM footprint**, 100% multicore **CPU utilization** (any scale-up) and **distributed execution** (scale-out), implemented using state-of-the-art **C++14/17 template meta-programming**.
- Implemented **core DB features** and system optimizations using multicore algorithms and **cache-aware data structures**.
- Designed thread management solution **improving** query engine performance **by order of magnitude**.
- Optimized query operations (join, group) on multi-billion record datasets.
- Implemented statistical **algorithms from academic papers**.

Early years - Allot Communications, ZRRO Technologies

FIRMWARE / EMBEDDED ENGINEER

2011-2014

- Design & implementation of **real-time algorithms & data structures** in a resource-restricted, **performance-critical** environment.
- Network protocols, coding of algorithms & systems in **C & C++** using extreme **memory/cpu optimizations** on **Linux** os.

Technical Skills

- **Modern C++ (11/14/17/20):** Expert-level proficiency including STL, Boost, Intel TBB. Advanced template metaprogramming.
- **Python & Backend Development:** Production experience with microservices architecture, distributed systems (MongoDB, RabbitMQ, Kafka). Async programming, performance optimization. Docker, Kubernetes for containerized deployments.
- **AI & LLM Systems:** Hands-on development of agentic workflows on multi-agent frameworks (LangGraph, LangChain, openAI agents SDK) with MCP protocol integration. Active coursework & hands-on project development.
- **Database Systems & Internals:** Deep expertise in DB internals (query parsers, execution engines, optimizers, caching layers). Designed and built ETL engine from scratch. Working knowledge of PostgreSQL, MongoDB, ClickHouse. Query language DSL design.
- **Performance Engineering:** Low-latency algorithms, memory/CPU profiling, cache-aware data structures, multi-threading.
- **Additional Languages:** Java (Spring, Maven), C# (Azure microservices).
- **Cloud Platforms:** Azure services (Blobs, Queues, CosmosDB, KustoDB), AWS (EC2, S3, Lambda).
- **Big data analytics:** Implement Hadoop/Spark data pipelines for cross-organization ETL workflows.
- **Distributed graph systems:** Built solutions for on-prem and HPC environments, coordinating millions of tasks.

Education

The Open University of Israel

Ra'anana, Israel

B.SC IN MATHEMATICS AND COMPUTER SCIENCE

2007 - 2010

- Graduated **summa cum laude** with GPA of **97/100**.
- Awarded place on University President's Excellence List (annual GPA ≥ 95) for four consecutive years.
- Achievement recognized in university educational portal [article](#).

Independent Work & Technical Writing

AI Projects, Udemy AI Learning & Technical Writing

- **Technical C++ writing:** Published a series of articles on database implementation & C++ techniques on LinkedIn, republished by [EuropeClouds](#) (2020-2021). Created [C++ TMP Tutorial](#) repo referenced in articles.
- **LangGraph Infrastructure:** Created Python abstraction layer providing reusable components (nodes, edges, tools, structured outputs) for agent development. [GitHub](#).
- **Agentic Workflows:** Implement multiple projects (stock market predictor, real-time soccer betting system) utilizing my infra project & external MCP integrations. [Stocks](#), [Betting](#).
- **The Complete Agent & MCP Course** (Udemy) - completed a comprehensive training in AI agents platforms.